

DiscoverEat

Occasion-Based Restaurant Discovery Platform

Web Systems Development — Mini-Project Report

Student: Abdul Hakim NAZARI

mentor: assoc. prof. dr. Boštjan Šumak

University of Maribor

Academic Year: 2025/2026

Deadline: June 7, 2026

1. Problem Description

Finding the right restaurant is rarely just about food — it is about **the experience matching the occasion**. Existing platforms like Google Maps, TripAdvisor, or Yelp answer "*what is nearby?*" but fail to answer "*what is right for tonight?*" A business lunch requires completely different criteria than a romantic anniversary dinner or a casual friends hangout. Generic filters like cuisine type or price range do not capture social context, atmosphere, or vibe.

DiscoverEat solves this by placing the *dining occasion* at the center of search. Users select their occasion first — Date Night, Family Dinner, Business Meeting, Friends Hangout, or Special Celebration — and are matched with restaurants

curated specifically for that context. This reduces decision fatigue and improves satisfaction with dining choices.

2. Target Users

- **Primary — Diners:** Adults 20–45 who dine out regularly and want contextually relevant recommendations, not generic lists. Mobile-first users who make decisions quickly.
- **Secondary — Restaurant Owners:** Business owners who want to reach diners actively looking for their type of dining experience and occasion.

3. User Personas

Persona 1 — Sofia Kovac (The Social Planner)

Age	27	Occupation	Marketing Coordinator, Ljubljana
Tech Literacy	High	Device	iPhone 14 — uses apps daily

Goals

- Find restaurants that match a specific social vibe (casual, fun, lively)
- Compare a few options quickly to share with friends in a group chat
- Avoid reading irrelevant reviews for the wrong type of occasion

Pain Points

- Existing apps offer no way to filter restaurants by "atmosphere"
- Cannot tell from a listing whether a place suits a large group
- Reads 15 reviews before realizing the restaurant is too formal for her needs

"I just want to know if the vibe is right — not just the rating."

Scenario: It is Friday afternoon. Sofia needs a place for 8 friends tonight. She opens DiscoverEat, taps "Friends Hangout", and within 2

minutes has 3 perfect options. She shares the link directly to the group chat.

Persona 2 — Marco Bianchi (The Busy Executive)

Age	42	Occupation	Regional Sales Director, Maribor
Tech Literacy	Medium	Device	Samsung Galaxy S24 — productivity use

Goals

- Find quiet, professional venues suitable for client meetings
- Access menu, pricing, and hours quickly without reading long reviews
- Make or check reservations in minimal steps

Pain Points

- Cannot filter for "business-appropriate" atmosphere on existing apps
- Star ratings mix casual and formal contexts — unreliable for business use
- Frequently ends up in venues too loud or informal for client meetings

"I need a place that says we take this seriously without saying a word."

Scenario: Marco has a client from Vienna next Tuesday at 13:00. He selects "Business Meeting", picks a 4.9-rated venue nearby, checks the menu prices, and taps Make Reservation — all in under 3 minutes.

4. Requirements

4.1 Functional Requirements

ID	Requirement	Priority
FR-01	Browse restaurants without logging in (guest mode)	Must Have
FR-02	Filter by occasion type (Date Night, Family, Business, Friends, Special Events)	Must Have
FR-03	Search restaurants by name or keyword	Must Have
FR-04	Search restaurants by city or location	Must Have
FR-05	View full restaurant profile (cuisine, rating, hours, contact, menu, gallery)	Must Have
FR-06	Add and remove restaurants from Favorites (registered users)	Must Have
FR-07	Register a new account with email and password	Must Have
FR-08	Sign in and sign out of account	Must Have
FR-09	Edit user profile (name, email, phone, avatar)	Should Have
FR-10	Interactive Mapbox map showing restaurant location on detail page	Should Have
FR-11	Browse restaurant menu by category tabs (Starters, Mains, Desserts)	Must Have
FR-12	Photo gallery with lightbox full-screen view	Should Have
FR-13	Business portal for restaurant owners to manage listings	Could Have

FR-14	Sort restaurants by rating or name on Discover page	Should Have
FR-15	Auto-playing featured restaurant carousel on home page	Should Have

4.2 Non-Functional Requirements

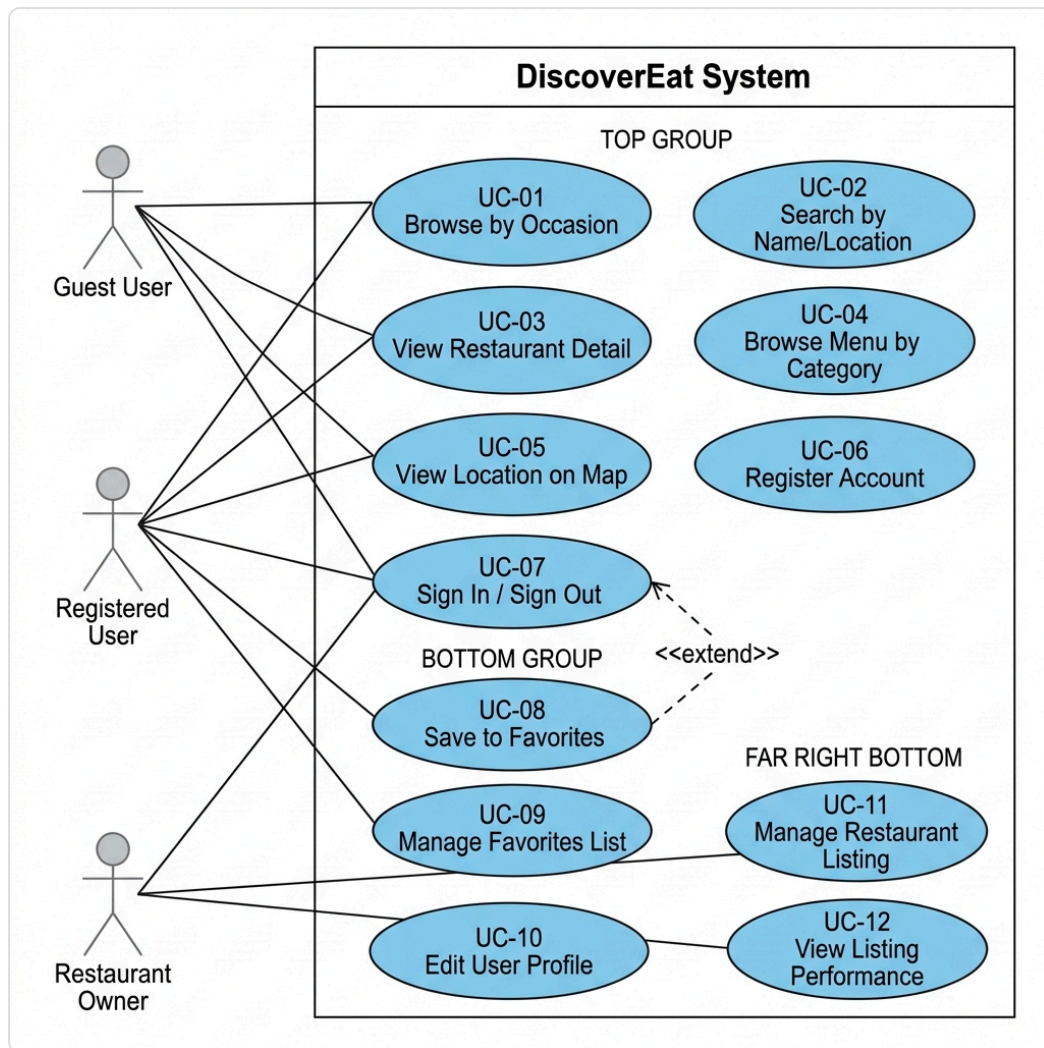
ID	Requirement	Category
NFR-01	Fully responsive across mobile, tablet, and desktop	Usability
NFR-02	Pages load within 3 seconds on standard broadband	Performance
NFR-03	UI honors system light/dark mode preference	Usability
NFR-04	All interactive elements have visible hover and focus states (WCAG)	Accessibility
NFR-05	Works in Chrome, Firefox, Safari, and Edge (latest versions)	Compatibility
NFR-06	Credentials handled securely — no plaintext passwords stored	Security
NFR-07	All components typed with TypeScript, no any types	Maintainability
NFR-08	Semantic HTML5 for SEO and screen reader compatibility	Accessibility
NFR-09	Animations complete within 300ms without layout shift	Performance

5. Use Case Model

5.1 Actors

Actor	Description
Guest User	Visitor without account — can browse, search, and view restaurant details
Registered User	Account holder — can save favorites, manage profile
Restaurant Owner	Business account — manages listing via the For Business portal
System	The DiscoverEat platform — handles filtering, routing, and data display

5.2 Use Case Diagram



5.3 Key Use Case Descriptions

UC-01: Browse Restaurants by Occasion

Actor	Guest User / Registered User
Precondition	User is on the Home or Discover page
Main Flow	1. User sees 5 occasion categories. 2. User clicks an occasion card. 3. System filters restaurants tagged with that occasion. 4. Matching restaurant cards are displayed.
Alt. Flow	No restaurants match → system displays empty state message
Postcondition	Filtered, relevant restaurant list is shown

UC-03: View Restaurant Detail

Actor	Guest User / Registered User
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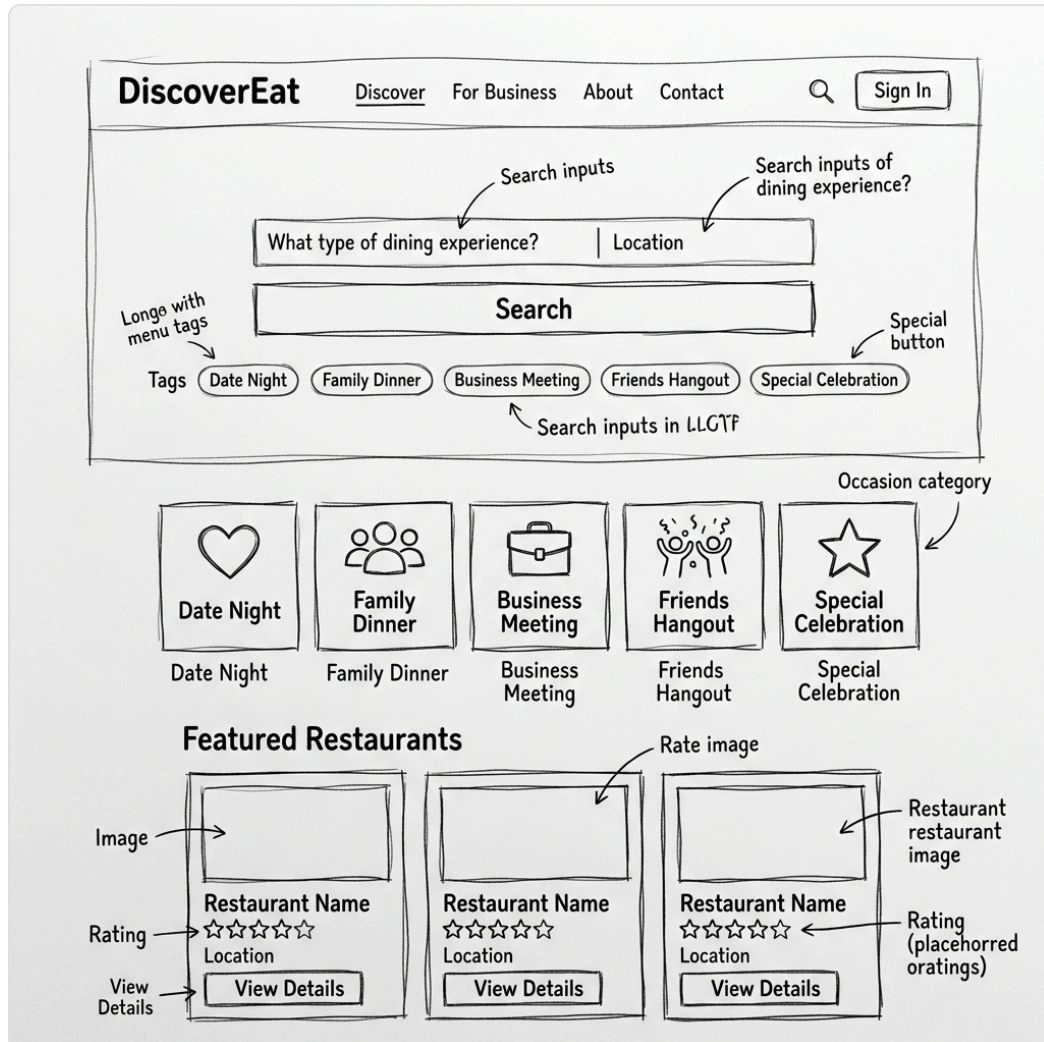
Precondition	User sees a restaurant card in results or Discover page
Main Flow	1. User clicks View Details. 2. System routes to /restaurant/:id. 3. Page shows hero carousel, rating, cuisine, price. 4. User browses tabbed menu (Starters/Mains/Desserts). 5. User views gallery with lightbox. 6. Sidebar shows address, hours, phone, website. 7. Map toggled on demand.
Postcondition	User has complete information to make a dining decision

UC-o8: Save Restaurant to Favorites

Actor	Registered User
Precondition	User is signed in and viewing a restaurant detail page
Main Flow	1. User clicks Heart icon. 2. System sets isFavorite = true. 3. Heart fills red (microinteraction). 4. Restaurant added to /favorites.
Alt. Flow	Not signed in → redirect to /signin with return URL
Postcondition	Restaurant is saved to user favorites list

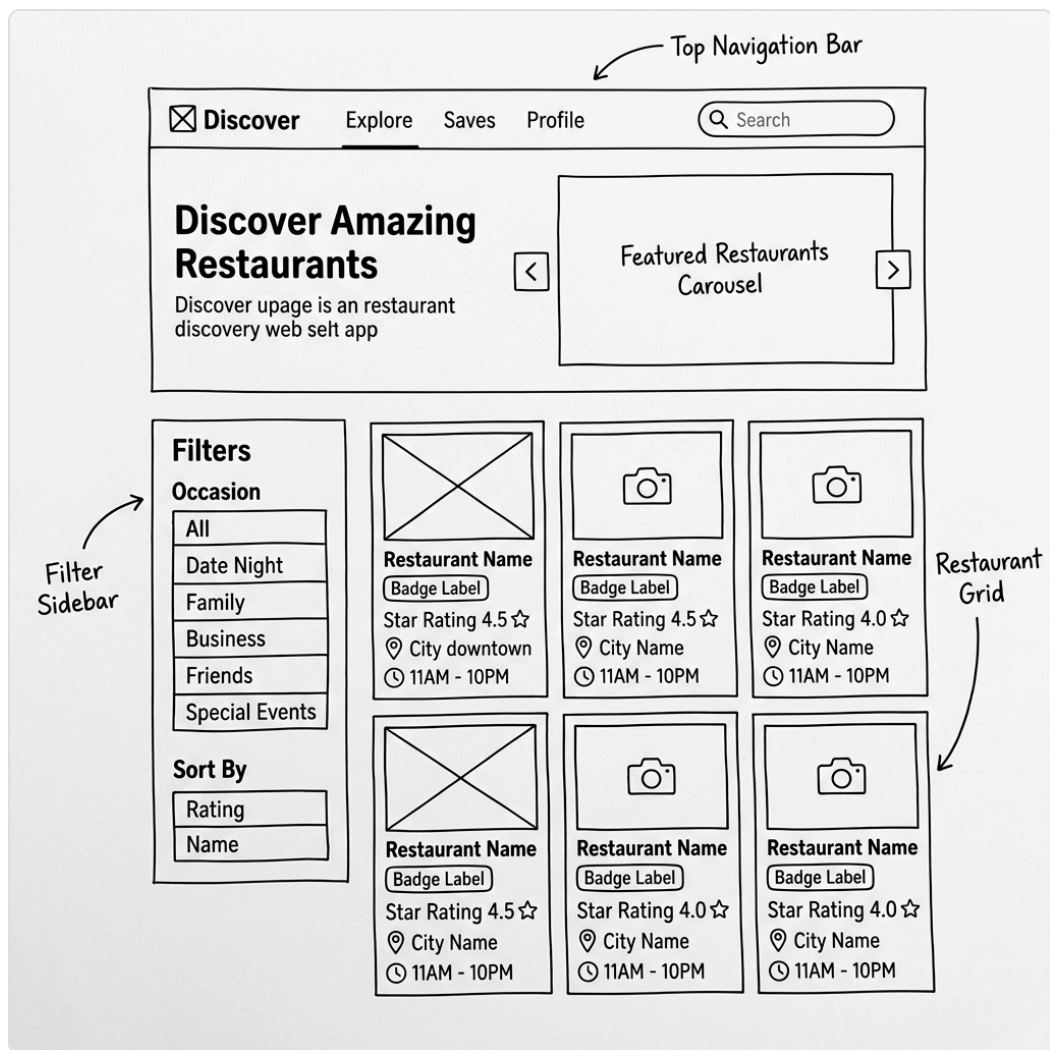
6. Wireframes

6.1 Home Page Wireframe



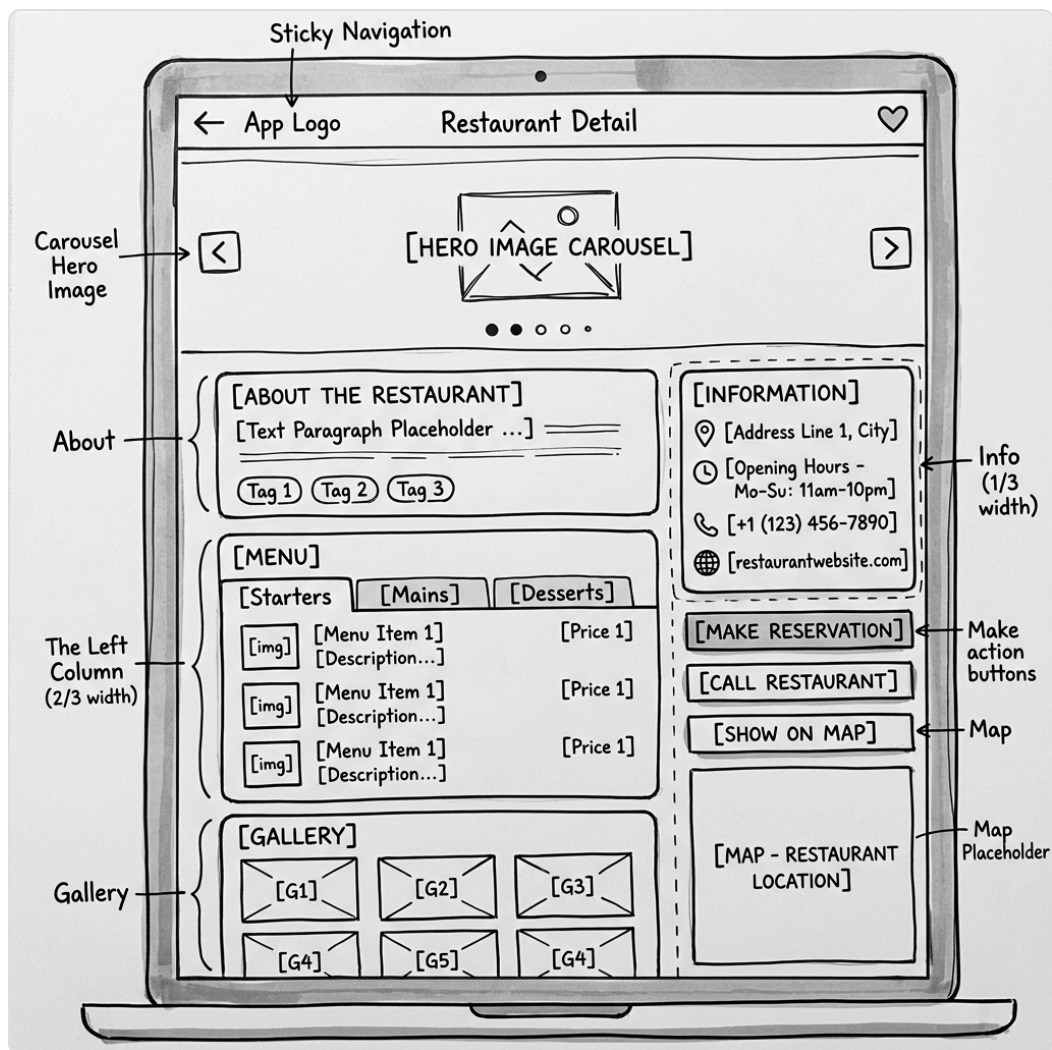
Key decision: Search bar and occasion shortcuts merged into a single hero section. Originally planned on separate pages — combined for one-step discovery without navigation.

6.2 Discover Page Wireframe



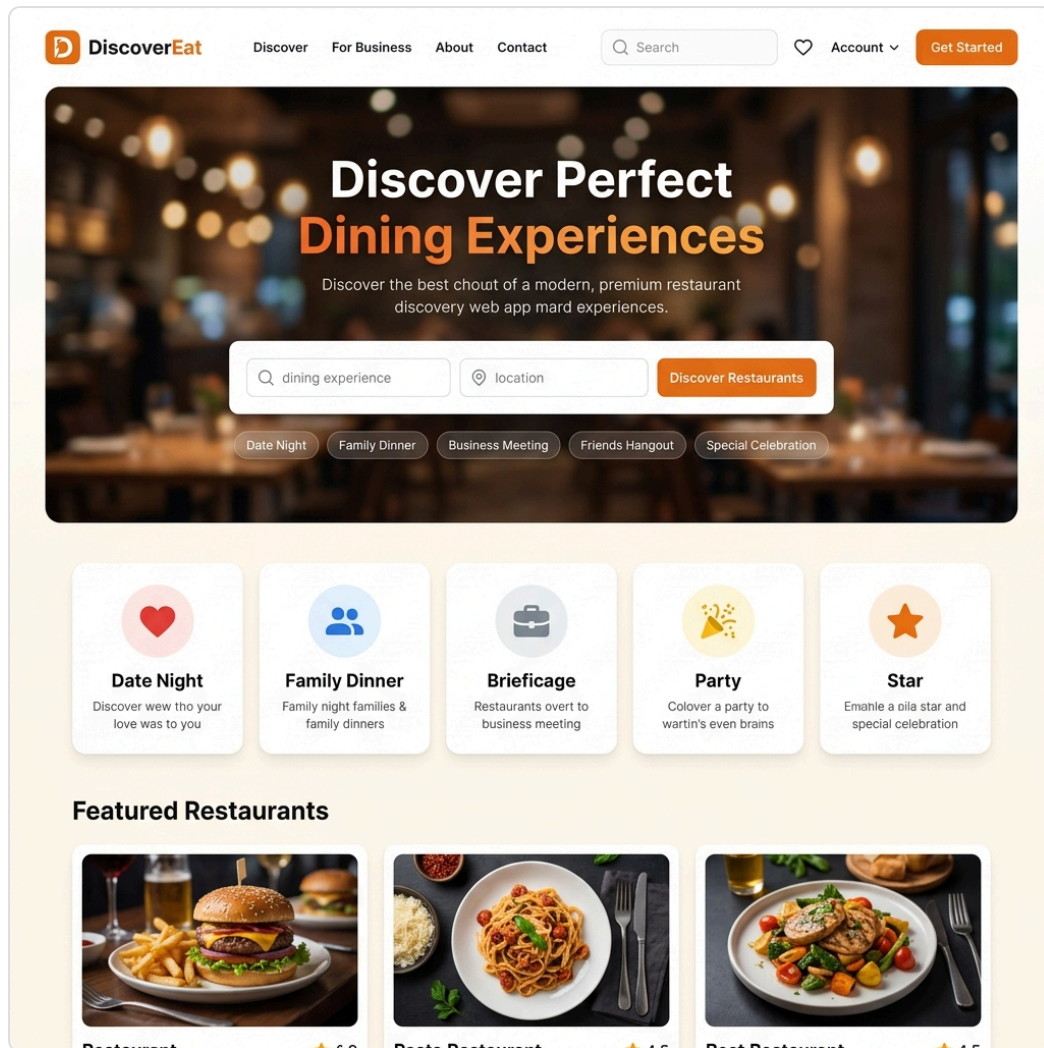
Key decision: Filter sidebar made persistent (not a dropdown) to encourage exploration. Featured carousel placed at top before the grid.

6.3 Restaurant Detail Wireframe



Key decision: Two-column layout separates browsable content (left) from quick-action sidebar (right). Map hidden by default, toggled on demand.

7. High-Fidelity Mockup

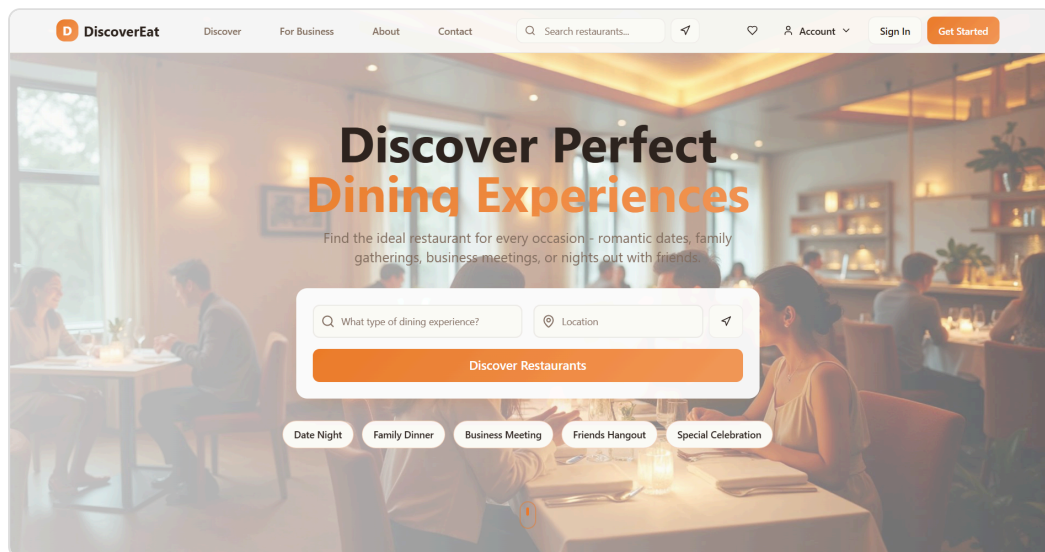


- **Primary color:** Warm orange #E8791D — evokes appetite, warmth, and energy
- **Hero:** Full-screen restaurant photo with gradient overlay for text legibility
- **Headline:** Two-tone 7xl bold heading (dark charcoal + orange gradient on key phrase)
- **Search card:** White floating card with soft shadow lifts the CTA above the background
- **Occasion tags:** Pill-shaped outlined buttons with orange hover state

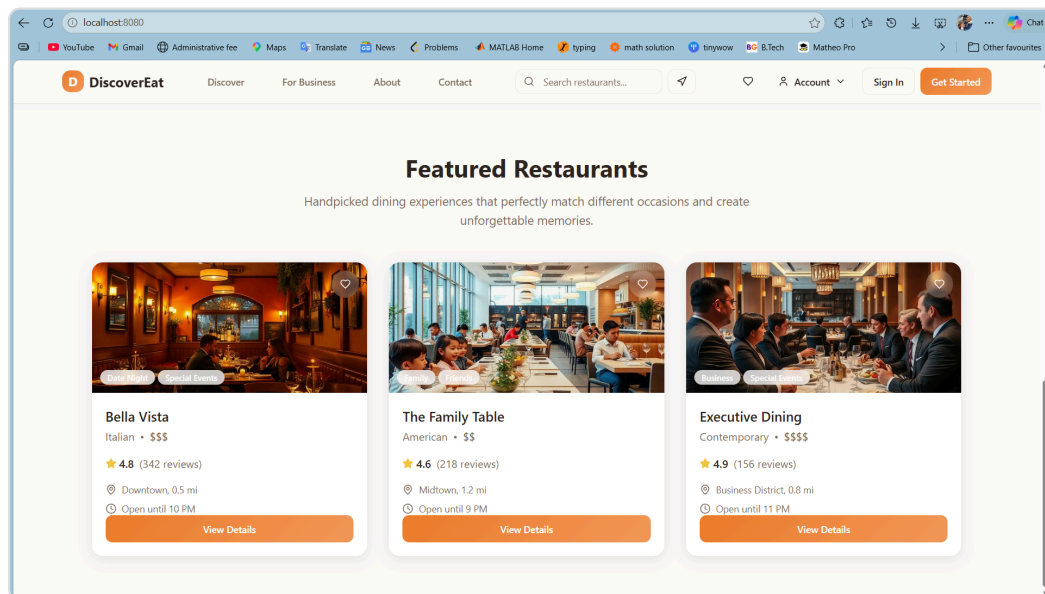
8. Interactive Prototype — Screenshots

The interactive prototype is a fully functional React 18 application running at **http://localhost:8080**, built with TypeScript, Vite, Tailwind CSS, shaden/ui, and Mapbox GL.

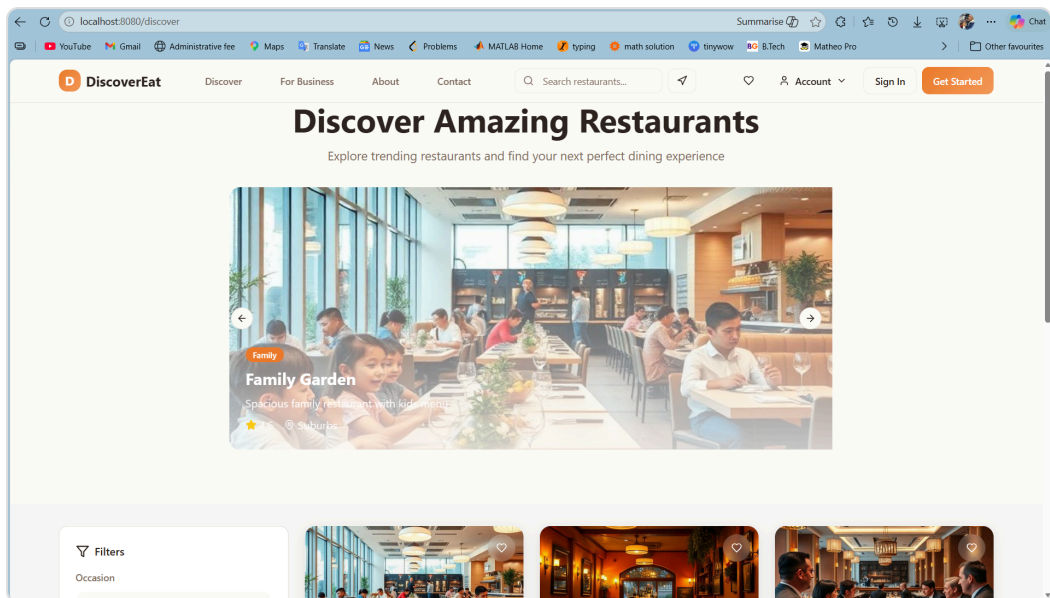
Home Page — Hero Section



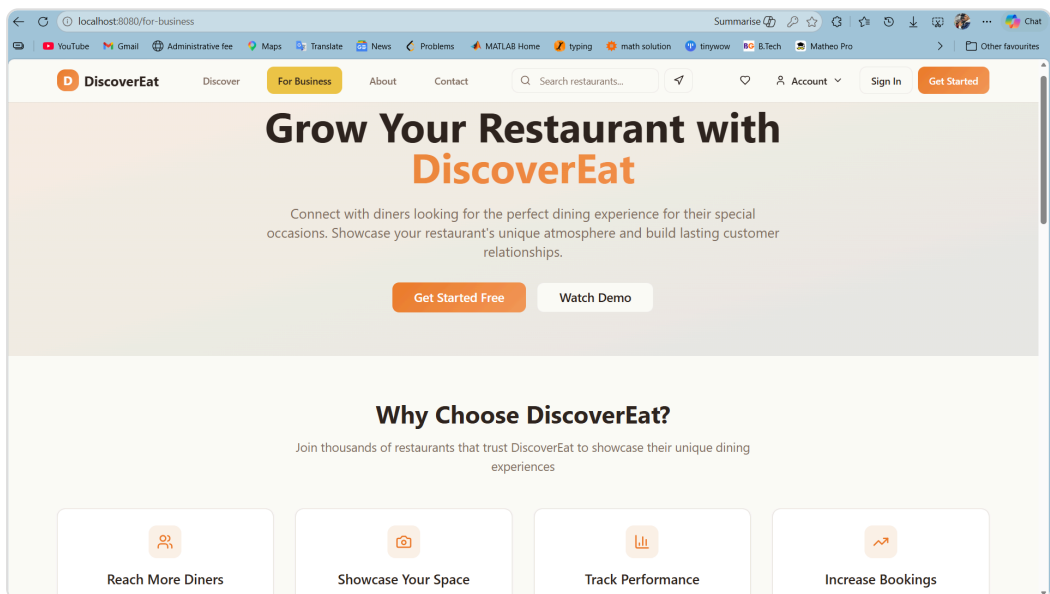
Featured Restaurants Section



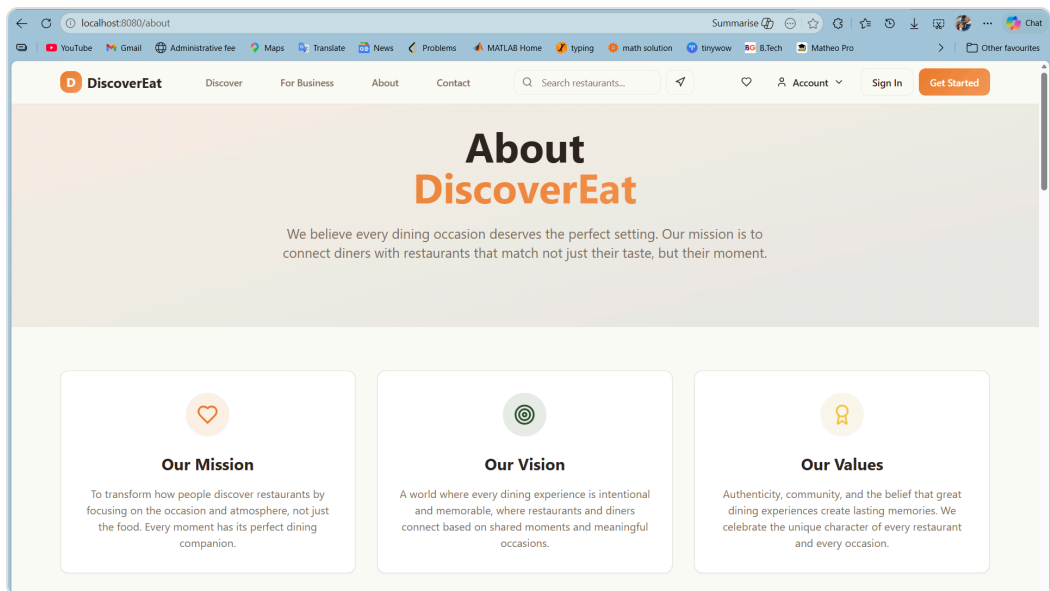
Discover Page — Browse and Filter



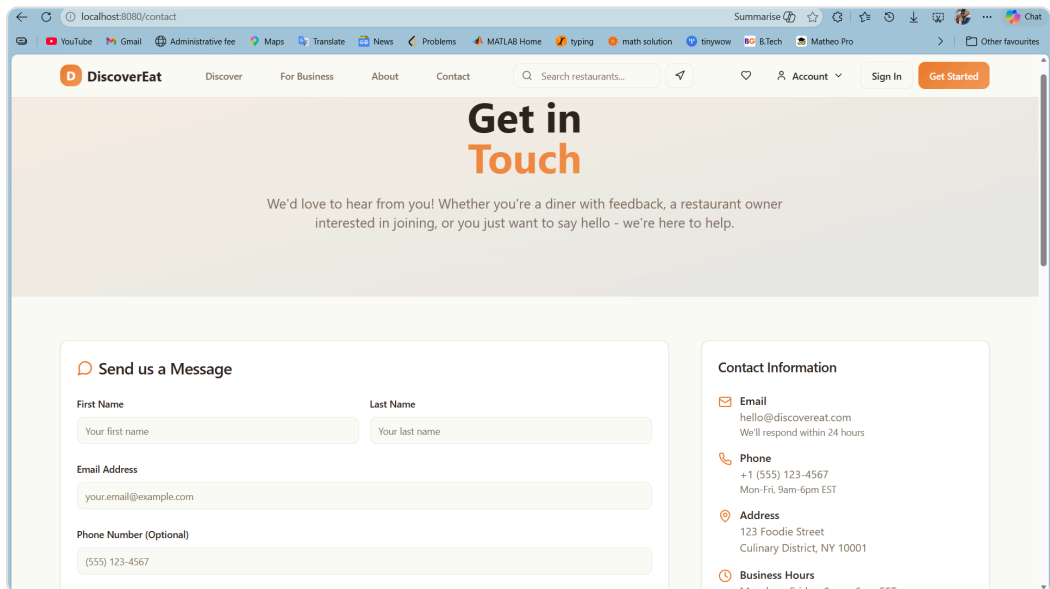
For Business Portal



About Page



Contact Page



9. Design Trends

Trend 1: Responsive / Mobile-First Design

Why selected: Restaurant discovery is inherently mobile. Both personas use phones: Sofia books on her iPhone during lunch breaks; Marco checks options on his Galaxy while commuting. A design that fails on mobile fails entirely.

How implemented: Tailwind CSS mobile-first breakpoints (md:, lg:). Restaurant card grids switch from 3-col desktop to 1-col mobile. Navigation collapses on small screens. Occasion filter changes from 5-col to 1-col. All inputs are touch-optimized.

How it improves UX: Every user gets a seamless experience regardless of device. The core use case — finding a restaurant on the go — is never compromised.

Trend 2: Microinteractions

Why selected: Food discovery is emotional. The interface must reflect the same care and quality it promises in the restaurants it showcases. Static, unresponsive UIs feel untrustworthy.

How implemented: Heart icon fills red on click (favorites toggle). Occasion cards scale with hover:scale-105. Hero scroll indicator bounces with animate-bounce. Card shadows elevate on hover:shadow-md. Featured carousel auto-plays at 3.5s intervals via Embla Carousel.

How it improves UX: Every interaction receives immediate visual acknowledgment. Users never wonder if their action registered, reducing confusion and building confidence in the interface.

Trend 3: Bold Typography and Visual Hierarchy

Why selected: Food experiences are emotional and aspirational. The copy must feel inviting and confident, not clinical. A bold 7xl headline creates an immediate impression of quality matching the premium experiences the platform represents.

How implemented: Hero uses text-4xl md:text-6xl lg:text-7xl font-bold. The key phrase uses bg-gradient-primary bg-clip-text text-transparent. Three-level hierarchy: Display (hero) → Heading (sections) → Body (supporting text).

How it improves UX: Clear hierarchy lets users scan the page in 2–3 seconds. Page purpose, section titles, and key actions are instantly identifiable without reading everything.

10. Design Evolution

Phase 1 — Initial Concept

Starting idea: Restaurant search filtered by food type (Italian, Asian, etc.) and star rating.

Problem discovered: Functionally identical to Google Maps, TripAdvisor, and Yelp — no genuine differentiation.

Key decision: Pivot entirely to occasion-based filtering. Insight: people choose by the type of evening they want to have, not cuisine first.

Phase 2 — Wireframes

Screens wireframed: Home, Discover, Restaurant Detail, Sign In/Register, User Profile, For Business Portal.

Problem discovered: Search bar on Home page and occasion filters on a separate Discover page — users had to navigate twice before seeing relevant results.

Key decision: Merge search input and occasion tags into a single hero section for one-step discovery from the landing page.

Phase 3 — Visual Design and Mockup

Decisions made: Adopted shadow/ui for accessible, consistent components. Chose warm orange #E8791D as brand color (food psychology: orange stimulates appetite and suggests warmth). Full-screen hero photography with gradient overlay. Occasion filter redesigned from text links to large interactive icon cards.

Problem discovered: Restaurant detail page was text-heavy. Users could not quickly assess if a place suited their occasion.

Key decision: Added photo carousel at the top, reorganized menu into category tabs, moved contact info to a sticky sidebar.

Phase 4 — Interactive Prototype

Built with: React 18, TypeScript, Vite, Tailwind CSS, shadcn/ui, Mapbox GL, React Router v6.

Improvements discovered during development:

- Added animated scroll indicator after users did not scroll past the full-height hero section
- Moved occasion tags into the hero after users expected them immediately on first load
- Changed filter from a dropdown to a persistent sidebar — makes filtering feel like exploration, not a task
- Added photo lightbox after observing users wanting to examine food images more closely
- Map hidden by default and toggled on demand to avoid unnecessary Mapbox API calls on page load

11. Summary

DiscoverEat addresses a genuine gap in the restaurant discovery market: the absence of occasion-aware, contextually relevant recommendations. By centering the user experience around the social purpose of the dining event, the platform creates a fundamentally more relevant and satisfying experience for its core users.

The application was developed through a structured design process: problem definition → user personas → requirements → use case modeling → wireframes → mockup → interactive prototype. At each stage, concrete problems were identified and deliberately resolved. Three modern design trends — responsive/mobile-first design, microinteractions, and bold typography — were selected because they directly serve the mobile usage context and the emotional nature of dining experiences.

